31/08/2019:

Continuous Delivery

(Artifactory / jfrog)

Introduction: (Before JFrog, nexus)

① Artifactory is a repository manager created by JFrog.

② A repository manager allows to store and retrieve build artifacts (.war / .jar)

③ A repository manager is a dedicated server application designed to manage binary components for the application that we build.

④ JFrog Artifactory is the only Universal Repository Manager supporting all major packaging formats build tools and CI servers.

⑤ JFrog provides both an open source and paid support version of the Artifactory Maven repository software.

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⑥ GitHub is used for source code and Artifactory is used for the binaries which go with the source code.

Jfrog Artifactory features

① Reliability

② Efficiency

③ Security (who can access / where to deploy)

④ Stability

⑤ Automation (using REST API)

Binary
Component
.war files
artifacts.
Same

Jfrog Artifactory (Night builds)

① On UAT (clean builds)

② Every night 2AM

③ Build periodically

④ Run maven target clean install

⑤ War file sent to remote artifactory by using jfrog.

⑥ Functional / performance test cases

⑦ Execute code analysis by using sonarqube (optional)

⑧ Deploy the artifact into UAT

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Jfrog artifactory installation Steps:

1. Launch amazon Linux machine with t2.medium instance type
2. While launching. add a rule 8081 in security groups
3. Install Java 8 (Removing default java 7)
4. Google search jfrog and click on first link.
5. Select ON-Prem Trial

↓

free trial

↓

I need integration with (select 5)

↓

please contact for neither

↓

Provide contact details.

↓

click on proceed.

↓

A license key is sent to Gmail.

↓

open gmail

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↓

Right click on link (first one)

↓

Select RPM & see the below steps:

↓

copy the first step and paste in the linux machine ↳

↓

Second step ↳

↓

third step ↳

↓

Installation successfully completed

↓

→ service artifactory check
(checking the installation status)

↓

→ service artifactory start
(starting the service)

↓

→ service artifactory status
(checking the status)

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03/09/2019:

① Enter the default username and password like below:

→ Username : admin

→ Password : password

② click on next.

③ Copy the license from Gmail account and paste in jfrog artifactory browser and click on next.

④ Click on skip (twice)

⑤ click on finish.

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⑥ Under admin dropdown, select local repository and select package type maven

⑦ Enter repository key (any name but unique)

⑧ Click on save and finish

⑨ Open jenkins dashboard click on manage jenkins

⑩ click on manage plugins, available tab

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(11) In filter, type artifactory → install the Artifactory plugin and restart the jenkins.

(12) Now: manage jenkins
↓
configuration systems.

(13) Under artifactory configuration:

Server id : <any name>

URL : <artifactory url>

(https://<publicip>:8081/artifactory)

(14) Username: admin
Password: password.

(15) click on test connection to check jenkins server is communicating with jfrog artifactory url or not.

(16) click on apply & save.

(17) Now in jenkins, create a job
↓
freestyle
↓
ok → SCM → Git

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(18) copy the repo URL & paste in the jenkins.

(19) Click on Build Environment tab and select the generic artifactory integration checkbox

(20) Select legacy pattern and click on refresh repositories.

(21) Select the repository key in target repository dropdown which we have given in artifactory

(22) In published artifacts enter the

**/*.war ⇒ ${BUILD_NUMBER}

Jenkins variables.

(23) click on build tab

↓

Invoke top level maven targets.

(24) Clean install in goals field.

↓

Build now.

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(25) After success, goto jfrog artifactory browser & refresh

(26) click of repository key, we can find the war files.

"CI/CD Workflow:"

(1) Based on the requirement, our developer will build the code and place it in feature branch

(2) Once it is done, it will go to the reviewer branch. Here one of the technical lead will review the code

(3) After that it will go to the manager approval branch, once it gets manager approval, we will merge with master.

(4) After that we will take a copy of code and put in release branch

(5) After that we will integrate git with jenkins by using git plugin/git webhooks

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(6) Once it is done, we will create a job like freestyle job, maven job, pipeline, multipipeline jobs.

(7) While creating jobs, we will provide job description

(8) Here, we have to provide Git URL and git branch details.

(9) After that build trigger here we will provide either build periodically or poll scm option.

(10) After that build here we will provide pom.xml to get binaries and libraries of the particular project.

(11) Then, after post build actions, here we will configure domain mail ids of the developer to get update their builds.

(12) After that we will configure the artifactory repository i.e jfrog.

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(13) Based on the configuration of the artifacts we will build, it will hosted on the artifactory repository.

(14) These artifacts, we will deploy into web servers and Application servers by using configuration management tool i.e Ansible.

(15) This is how we have automated our CI CD process.

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